



PRECURSOR ANALYSIS REPORT: UKRAINE ENERGY SECTOR CYBER ATTACK 2015

Cybersecurity for the Operational Technology
Environment (CyOTE)

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TABLE OF CONTENTS

1. EXECUTIVE SUMMARY	2
2. INTRODUCTION	4
2.1. APPLYING THE CYOTE METHODOLOGY	4
2.2. BACKGROUND ON THE ATTACK	6
3. OBSERVABLE AND TECHNIQUE ANALYSIS	8
3.1. SPEARPHISHING ATTACHMENT TECHNIQUE (T0865) FOR INITIAL ACCESS	8
3.2. USER EXECUTION TECHNIQUE (T0863) FOR EXECUTION	10
3.3. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION	11
3.4. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL	12
3.5. NETWORK CONNECTION ENUMERATION TECHNIQUE (T0840) FOR DISCOVERY	13
3.6. MASQUERADING TECHNIQUE (T0849) FOR EVASION	14
3.7. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT	15
3.8. EXTERNAL REMOTE SERVICES TECHNIQUE (T0822) FOR INITIAL ACCESS	16
3.9. UNAUTHORIZED COMMAND MESSAGE TECHNIQUE (T0855) FOR IMPAIR PROCESS CONTROL	17
3.10. REMOTE SERVICES TECHNIQUE (T0886) FOR INITIAL ACCESS	18
3.11. GRAPHICAL USER INTERFACE TECHNIQUE (T0823) FOR EXECUTION	19
3.12. MANIPULATION OF CONTROL TECHNIQUE (T0831) FOR IMPACT	20
3.13. DENIAL OF CONTROL TECHNIQUE (T0813) FOR IMPACT	21
3.14. SYSTEM FIRMWARE TECHNIQUE (T0857) FOR IMPAIR PROCESS CONTROL	22
3.15. BLOCK COMMAND MESSAGE TECHNIQUE (T0803) FOR INHIBIT RESPONSE FUNCTION	23
3.16. BLOCK REPORTING MESSAGE TECHNIQUE (T0804) FOR INHIBIT RESPONSE FUNCTION	24
3.17. LOSS OF CONTROL TECHNIQUE (T0827) FOR INHIBIT RESPONSE FUNCTION	25
3.18. DENIAL OF SERVICE TECHNIQUE (T0814) FOR INHIBIT RESPONSE FUNCTION	26
3.19. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION	27
3.20. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION	28
3.21. DEVICE RESTART/SHUTDOWN TECHNIQUE (T0816) FOR INHIBIT RESPONSE FUNCTION	29
APPENDIX A: OBSERVABLES LIBRARY	31
APPENDIX B: ARTIFACTS LIBRARY	36
APPENDIX C: OBSERVERS	50
REFERENCES	51

FIGURES

FIGURE 1. CYOTE METHODOLOGY	4
FIGURE 2. INTRUSION TIMELINE	6
FIGURE 3. ATTACK GRAPH	30

TABLES

TABLE 1. TECHNIQUES USED IN THE UKRAINE ENERGY SECTOR CYBER ATTACK 2015	7
TABLE 2. PRECURSOR ANALYSIS REPORT QUANTITATIVE SUMMARY	7

PRECURSOR ANALYSIS: UKRAINE ENERGY SECTOR CYBER ATTACK 2015

1. EXECUTIVE SUMMARY

The Ukraine Energy Sector Cyber Attack 2015 Precursor Analysis Report leverages publicly available information about the cyber attack mounted against Ukraine's energy sector in December 2015 and catalogs anomalous observables for each technique employed in the attack. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

Ukraine was the victim of a highly sophisticated and coordinated cyber attack on 23 December 2015 that compromised three power distribution utilities, known as oblenergos, and allowed the adversaries to open the breakers at over 50 electrical substations. The attack caused a loss of power to approximately 225,000 customers for several hours.¹

The groundwork for the attack was laid as far back as May 2014, when spearphishing campaigns likely provided adversaries with initial access to the oblenergos' networks.² The adversaries employed an extensive array of malware and techniques in the course of their campaign: BlackEnergy, sshbear, and Kryptic malware; compromise of Virtual Private Networks (VPN) connected to a Distribution Management System (DMS) and Human-Machine Interface (HMI); and a Telephonic Denial-of-Service (TDoS) attack and adversary-scheduled outage of the Uninterruptable Power Supply (UPS) for a Private Branch Exchange (PBX) telephone system that served the oblenergos, which seriously degraded the ability of responders to communicate after the attack was triggered. While the impact on communications temporarily degraded the oblenergos' situational awareness after operators took the SCADA systems offline, they were able to restore power in six hours by sending engineers to manually control the substations.³

The adversary was able to control the victims' systems and disrupt their response without causing excessive damage to either the Supervisory Control and Data Acquisition (SCADA) systems or the electrical power grid itself. The main elements that were effectively destroyed were Serial-to-Ethernet converters rendered inoperable by a malicious firmware update and hard drives destroyed with KillDisk, which targeted the Master Boot Record (MBR) of victim machines, destroying just enough data to render systems inoperable during the attack.

Even though the attack did not result in catastrophic damage to the grid infrastructure, the oblenergos were operating the substations manually over two months later, still unable to remotely operate the substation breakers.⁴

Researchers and analysts identified 21 techniques utilized during the attack with a total of 86 observables using MITRE ATT&CK® for Industrial Control Systems. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If stakeholders perceive and investigate observables accompanying the attack techniques prior to the triggering event, earlier comprehension of malicious activity can take place. Ten of the identified techniques used during the Ukraine 2015 cyber attack were precursors to the triggering event. Case study analysts identified 48 observables associated with precursor techniques. CyOTE researchers assessed 34 observables to have an increased likelihood of perception during the 575 days preceding the triggering event.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references to support the comprehension of

indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

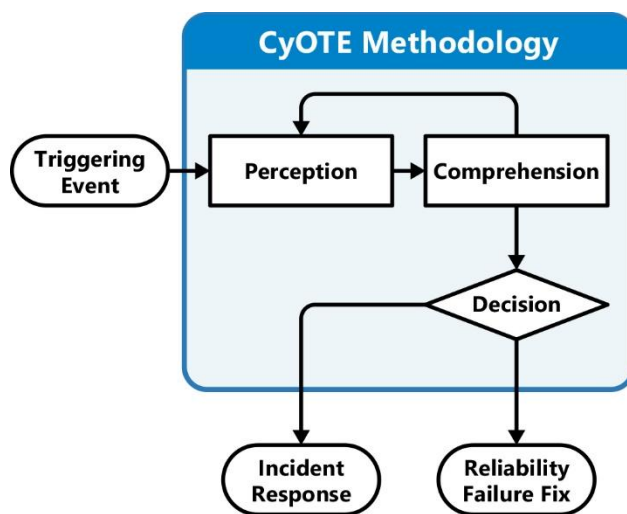


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the case study's cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references to support the comprehension of indicators of attack.

2.2. BACKGROUND ON THE ATTACK

Three Ukrainian power distribution utilities, known as oblenergos, were the victims of a cyber attack on 23 December 2015 (D-0) that resulted in a loss of power for approximately 225,000 customers.⁵

Spearphishing campaigns likely provided the adversaries with initial access as far back as May 2014 (D-575).⁶ Using harvested credentials, the adversaries launched a coordinated attack to open breakers at over 50 substations across the three oblenergos, causing an outage that lasted six hours.⁷

A timeline of adversarial techniques is shown in Figure 2. The timeline includes the estimated number of days prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.

At about 3:30 PM on 23 December 2015 (D-0), operators at Chernivistioblenergo noticed the cursors on their Human-Machine Interfaces (HMIs) moving on the screen (phantom mouse) to open their substation breakers.^{8,9} At roughly the same time, adversaries attacked Kyivoblenergo by sending commands via a rogue client directly to the Distribution Management System (DMS) to open breakers.^{10,11} Approximately 30 minutes later at Prykarpattyoblenergo, adversaries used the phantom mouse again to open that utility's substation breakers. The three victims had little time to respond once the adversaries triggered these attacks

Adversaries also anticipated and impaired the oblenergos' ability to respond to the outage. The adversaries overwrote firmware on the Serial-to-Ethernet converters that enabled control center operators to remotely control breakers (H-0 through H+1), forcing engineers into the field to take manual control of the substations. A coordinated Telephonic Denial-Of-Service (TDoS) attack disabled Kyivoblenergo's call center for three hours (H+3), preventing communication with customers. The adversaries also impacted key data centers and phone systems by scheduling Uninterruptable Power Supply (UPS) systems to go offline prior to the attack.¹² Finally, a wiper deleted the Master Boot Record (MBR) of systems that remained online, including those tied to finance and human-resources, rendering them useless and further adding to the chaos.

Full comprehension of the attack did not occur until two months later (D+60), when the Department of Homeland Security (DHS) issued a formal report on the event.¹³

Analysts and researchers identified 21 techniques in a sequence and timeframe likely used by adversaries during this cyber attack (Figure 3). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

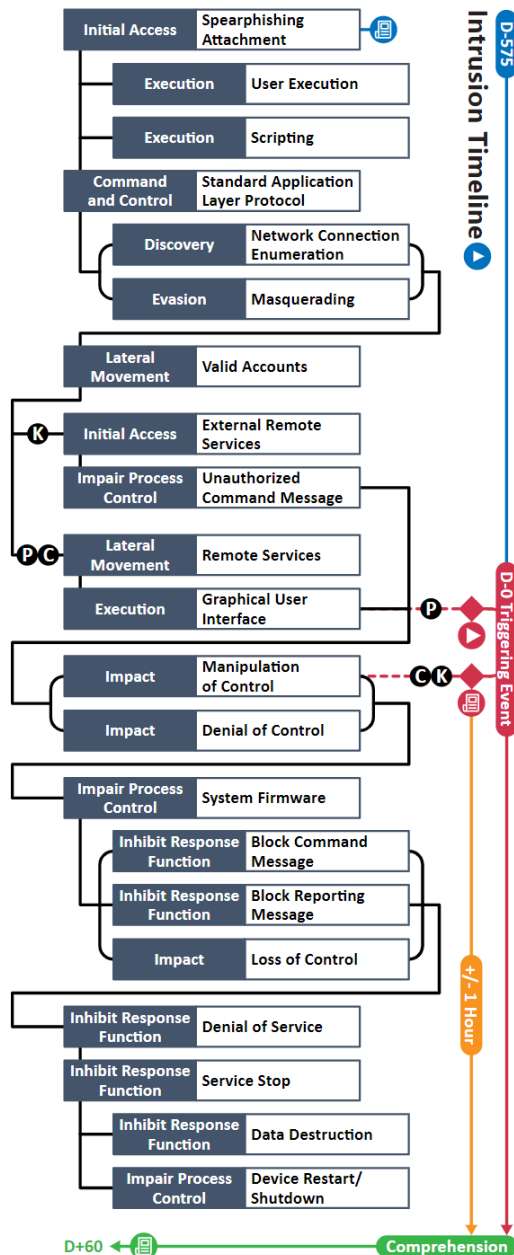


Figure 2. Intrusion Timeline

Table 1. Techniques Used in the Ukraine Energy Sector Cyber Attack 2015

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Wireless Compromise									System Firmware		

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	21
Technique Observables	86
Precursor Techniques	10
Precursor Technique Observables	48
Highly Perceivable Precursor Technique Observable	34

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. SPEARPHISHING ATTACHMENT TECHNIQUE (T0865) FOR INITIAL ACCESS

Adversaries used the Spearphishing Attachment technique (T0865) for Initial Access to communications networks of multiple companies within the Ukrainian energy sector. Adversaries sent malicious Word/Excel docs with embedded Object Linking and Embedding (OLE) objects.¹⁴ Malicious document names included, Додаток1.xls, \$RR143TB.doc, and associated components such as vba_macro.exe.¹⁵ If a victim downloaded and executed such attachments, these attachments would generate anomalous network connections or downloads over HTTP(S).

Adversaries conducted multiple phishing campaigns against the Ukrainian energy sector leading up to the December 2015 attack.^{16,17,18} On 12 May 2014, Prykarpattiaoblenergo received an email as part of a campaign targeting Ukraine's State Administration of Railway Transport. At the end of March 2015, an oblenergo in Western Ukraine was the target of a phishing attack.¹⁹ The spearphishing campaigns included forged sender addresses that appeared to impersonate at least two Ukrainian government entities, the Ukrainian Ministry of Energy and the Supreme Council of Ukraine.^{20,21,22}

Kryptic and Dropbear backdoors were discovered on hosts in Prykarpattiaoblenergo and Chernivtsioblenergo, respectively, and along with BlackEnergy 2/3 – a highly modular trojan – are additional possible access vectors for the December 2015 attack.^{23,24}

In addition, spearphishing attachments used in a subsequent attack in January 2016 initiated a download of a GCat binary compiled by PyInstaller.²⁵ When executed, this binary sends unusual SMTP traffic to Google or another webmail provider. File signatures extracted from network traffic or host data provide additional observables for BlackEnergy 2/3 or GCat.

Considering the prevalence of spearphishing attacks against Ukrainian targets, particularly the energy sector, CyOTE analysts assess that spearphishing was a likely access vector for the December 2015 attack against the three oblenergos.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff could perceive these reported pieces of evidence. This technique is important because it presents perceivable effects that, if identified and investigated earlier, could have reduced comprehension time by as much as 19 months. Terminating the chain of techniques at the Spearphishing Attachment technique (T0865) would likely prevent adversaries from accessing internal networks and delivering malicious payloads.

A total of seven observables were identified with the use of the [Spearphishing Attachment technique \(T0865\)](#). Five are assessed as highly perceivable (Object Linking and Embedding (OLE) Objects in Malicious Microsoft Office Documents; Додаток1.xls and Associated Components: vba_macro.exe; \$RR143TB.doc and Associated Components: vba_macro.exe; Unusual SMTP Traffic to Google or Other Web Mail Provider; File Signatures in Files Extracted from Network Traffic or Host Data).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 30 artifacts could be generated by the Spearphishing Attachment technique
Technique Observers^a	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

^a Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. USER EXECUTION TECHNIQUE (T0863) FOR EXECUTION

Adversary malware employed the User Execution technique (T0863) for Execution by sending victims malicious Microsoft Office documents. Targeted victims would likely notice suspicious emails with spoofed sender addresses and attachments that prompt users to enable macros when downloaded. Once opened, these Word and Excel documents would spawn suspicious child processes associated with command prompt processes or network communications. As a result, HTTP or Server Message Block (SMB) protocols may contain known malicious signatures for Command and Control (C2) or malware propagation.

Victims of spearphishing campaigns likely provided one vector for initial access leading to the December 2015 power outages. Email recipients would have been prompted to download and open weaponized Microsoft Office files.²⁶ For example, the May 2014 spearphishing attack that affected Prykarpattiaoblenergo employed a Portable Executable (PE) file disguised as a Microsoft Word document.²⁷ In another example, the March 2015 phishing attack targeted grid operators in Western Ukraine via a weaponized Microsoft Excel file Додаток1.xls.²⁸ BlackEnergy malware, if it was being employed, would install once the email recipients enabled the malicious macros on their workstation.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff who monitor enterprise and operational networks, as well as endpoint devices, could perceive these reported pieces of evidence. IT Cybersecurity, and IT Staff would likely observe communications with external IP addresses via HTTP traffic, as well as malicious signatures associated with embedded scripts within Microsoft Office documents.²⁹ It is important to note, however, that identifying Indicators of Compromise (IoC) is not enough to stop infection. For example, in the May 2014 spearphishing campaign the executable PE file was detected by 16 of 52 antivirus products, but still resulted in an infection.³⁰

A total of three observables were identified with the use of the [User Execution technique \(T0863\)](#), all of which are assessed as highly perceivable (Office Documents Associated with Network Communications, Command Prompt Processes, or Other Suspicious Child Processes; HTTP or SMB Protocols Containing Known Malicious Signatures for C2 or Malware Propagation; Suspicious Emails with Spoofed Sender Addresses and Attachments that Prompt Users to Enable Macros When Downloaded).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the User Execution technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.3. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

Malicious Microsoft Office documents containing OLE (or another scripting object) execute and install malware on the victim workstation as part of the Scripting technique (T0853) for Execution. Child processes of these malicious MS Office documents then generate network traffic for C2 and scan the victim network. Malicious macros written in Visual Basic for Applications (VBA) install plugins on the email recipient's workstation containing libraries for keylogging, network scanning, taking screenshots, or stealing passwords.³¹

Files specific to the 2015 campaign included Додаток1.xls, \$RR143TB.doc, VBA_macro.exe, CPLEXE.EXE (original name), MS-IME (internal name), virus_04.exe, vba_macro, icshextobin.exe, BlackEnergy.exe, and 1.exe.^{32,33,34}

This technique would very likely have occurred immediately following the User Execution technique in Section 3.2 and, by inference, likely occurred sometime between May 2014 and March 2015. Execution of this technique would provide data from reconnaissance of the victim networks that would be used in later stages of the attack.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff who monitor enterprise and operational networks, as well as endpoint devices, could perceive these reported pieces of evidence. These observers would likely observe an increase in network traffic as well as the execution of specific services on endpoints.

A total of three observables were identified with the use of the [Scripting technique \(T0853\)](#), all of which are assessed as highly perceivable (Office Document Containing OLE or Other Scripting Object; Network Traffic Associated with Microsoft Office Processes; Filenames Specific to 2015 Campaign).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.4. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL

The Standard Application Layer Protocol technique (T0869) was used for Command and Control. The C2 traffic resulted in anomalous outbound network connections or downloads over HTTP(S).³⁵ This traffic was associated with a parent process for a malicious Microsoft Office document containing OLE or another scripting object. Communications to a C2 server also made an HTTP request directly to an IP with no associated DNS request.³⁶ If GCat, a backdoor trojan, was used, one could observe SMTP connections to unusual or unexpected mail services. In addition, C2 servers can encrypt some HTTP communications with the RC4 algorithm.³⁷

As early as May 2014 and as late as December 2015, the BlackEnergy malware deployed to victim networks likely established a remote connection with a C2 server via HTTP POST requests to download additional malware plugins.³⁸ These plugins, described in Section 3.3, could be used to collect information about the victim networks.^{39,40}

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff could perceive these reported pieces of evidence.

A total of four observables were identified with the use of the [Standard Application Layer Protocol technique \(T0869\)](#). Terminating the chain of techniques at this stage would limit operational damage and reconnaissance on victim systems. Of the four observables associated with this technique, three are assessed to be highly perceivable (Anomalous Outbound Network Connections/Downloads Over HTTP(S); Microsoft Office Documents Containing OLE or Other Scripting Objects as Parent Process Associated with HTTP Comms to External IP; SMTP Connections to Unusual or Unexpected Mail Services).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Standard Application Layer Protocol technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.5. NETWORK CONNECTION ENUMERATION TECHNIQUE (T0840) FOR DISCOVERY

Adversary malware employed the [Network Connection Enumeration technique \(T0840\)](#) for Discovery. Infected systems likely generated network traffic associated with scanning behavior including sequential ports, high destination to source port communication ratio, and high quantity of partial connections. The technique may also be associated with anomalous use of the MS Sysinternals PsExec tool.

Using malware such as BlackEnergy 3, the adversaries could map the network and scan hosts for open ports using modules such as scan.dll.^{41,42} This activity could have occurred as early as May 2014 at Prykarpattyaoblenergo and as late as December 2015 for the other victims. The three oblenergos used different DM), so adversaries needed to gather information to understand each of them.^{43,44} Once BlackEnergy operators obtained valid credentials, they could move to other computers via PsExec and the vs.dll module.⁴⁵ In the October 2015 attacks against Ukrainian media outlets, adversaries utilized this approach, which may have been used in December against the three oblenergos to enumerate and establish access to victim systems.⁴⁶

CyOTE analysts assess OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff monitoring operational networks and endpoint devices could perceive these reported pieces of evidence.

Of the of three observables associated with this technique, all are assessed to be highly perceivable (Network Traffic with Scanning Behavior from Unusual Systems; Filenames Specific to the 2015 Campaign; Anomalous Use of MS Sysinternals PsExec Tool).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 33 artifacts could be generated by the Network Connection Enumeration technique
Technique Observers	OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.6. MASQUERADING TECHNIQUE (T0849) FOR EVASION

The Masquerading technique (T0849) was used for Evasion. Adversaries began scanning the network with the BlackEnergy component scan.dll but then switched to the more effective nmap tool. This evidence suggests that adversaries had time to adapt their approach to conducting reconnaissance on victim networks.

Adversaries disguised the nmap tool as the svchost.exe executable in the directory C:\Windows\Temp\Syslog, while the legitimate svchost.exe is in C:\Windows\System32.⁴⁷ The illegitimate svchost.exe could be unsigned, have an invalid signature, have an unusual publisher signature, or a valid signature but invalid or unknown file hash. Defenders may observe unusual library imports and exports or application files within the operating system.

CyOTE analysts assess IT Cybersecurity and IT Staff could perceive these reported pieces of evidence.

A total of six observables were identified with the use of the [Masquerading technique \(T0849\)](#). If comprehended at this stage of the attack, the ability for the adversary to move through the network could be compromised depending on the number of systems previously infected. Two of the six observables are assessed as highly perceivable (Unsigned or Invalid Signature on Operating System or Application File; Unusual Publisher Signature on Operating System or Application File).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 14 artifacts could be generated by the Masquerading technique
Technique Observers	IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.7. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT

Adversaries employed the Valid Accounts technique (T0859) for Lateral Movement by gathering valid credentials to establish pathways into the control environments of the oblenergos on 23 December 2015.^{48,49} At 4:10 PM OT Staff responded to outages at one oblenergo by disabling an HMI administrator account, only to have the adversaries switch to another HMI account to continue the attack.⁵⁰ Pathways that enabled this capability included credentials for the Virtual Private Network (VPN) used to access the control network remotely and likely the Remote Desktop Protocol (RDP) service to access the Windows Domain Controller. Adversaries created additional unauthorized domain accounts in at least one instance.⁵¹ These credentials could have been harvested through BlackEnergy 3 plugins to log keystrokes, through compromise of the domain or by using mimikatz to retrieve Windows account passwords/hashes.⁵²

Suspicious or unusual logins to domain-joined assets observed in the Domain Controller or target machine's Windows Event Logs' Security Log and likely indicated anomalous use of valid accounts. SMB traffic associated with a user account that can invoke PsExec also indicates potential use of this technique to propagate across the network.⁵³ By default, PsExec creates a new service, named PSEXESVC, on the remote system that can be detected via Windows new service creation logs, Event ID 7045 and 4697. Unusual user account logins, particularly those associated with admin accounts across network services such as SMB, also indicate the Valid Accounts technique. As the BlackEnergy malware propagates, suspicious files such as ps.dll, ss.dll, and kl.dll are moved over SMB or other admin shares.⁵⁴

CyOTE analysts assess OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff could have perceived some of these observables through an investigation.

A total of eight observables were identified for the [Valid Accounts technique \(T0859\)](#). If identified and investigated earlier in the campaign, comprehension of this technique likely could have reduced impact by limiting compromise of assets on the victim network. Seven of the eight observables associated with this technique are likely highly perceivable (Suspicious or Unusual Interactive Successful Logins to Domain-Joined Assets Observed in the Domain Controller (Event 4624); Suspicious or Unusual Interactive Unsuccessful Logins to Domain-Joined Assets Observed in the Domain Controller (Event 4625); Traffic Associated with PsExec Due to User Account Compromise; Unusual User Account Logins, Particularly Admin Accounts Across Network Services such as SMB; PsExec Process Artifacts on the Destination System associated with PSEXESVC.exe; Filenames Specific to the 2015 Campaigns; Event ID 7045 and 4697).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Valid Accounts technique
Technique Observers	OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.8. EXTERNAL REMOTE SERVICES TECHNIQUE (T0822) FOR INITIAL ACCESS

The External Remote Services technique (T0822) was used for Initial Access. Adversaries gained access to oblenergo control networks as early as May 2014 with persistent access continuing through at least 23 December 2015 by using existing remote access tools similar to RDP and radmin.⁵⁵ Credentials for VPN access to oblenergo SCADA networks were likely obtained from Windows Domain Controllers, allowing the adversaries to interact directly with DMS clients, and the VPNs from the business network into the SCADA networks appeared to lack Two Factor Authentication (2FA).⁵⁶ Adversaries were then able to access the SCADA systems for Chernivtsioblenergo, Kyivoblenergo, and Prykarpattyaoblenergo at approximately 3:30 PM, 3:35 PM, and 4:06 PM, respectively, on 23 December.^{57,58}

As a result of this technique, VPN application logs likely showed anomalous account activity and VPN logon session metadata likely showed the use of a compromised account. Further, host logs contained unusual process data associated with remote access tools such as radmin.⁵⁹ These remote access connections could trigger network Intrusion Detection System/Intrusion Prevention System (IDS/IPS) alerts (e.g., snort rules).

CyOTE analysts assess OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff monitoring operational networks and endpoint devices could have perceived these observables through an investigation.

Four observables associated with the [External Remote Services technique \(T0822\)](#) were identified. Of the four observables associated with this technique, three are assessed as highly perceivable (VPN Application Logs of Anomalous Account Activity; VPN Logon Session Metadata Shows Use of Compromised Account; Network IDS/IPS Alerts Associated with radmin Connections).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 26 artifacts could be generated by the External Remote Services technique
Technique Observers	IT Staff, OT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.9. UNAUTHORIZED COMMAND MESSAGE TECHNIQUE (T0855) FOR IMPAIR PROCESS CONTROL

The Unauthorized Command Message technique (T0855) was used to Impair Process Control. One expression of this technique is to use anomalous VPN connections to access a key business application. In this case study, adversaries used a DMS client application to issue commands directly to a DMS server in the rogue client approach employed at Kyivoblenergo.⁶⁰ Even with the phantom mouse technique, operator HMIs reflected unauthorized process change inputs via the radmin log. Other observables associated with this technique in an operational environment include PCAP files showing unauthorized instructions sent to field controllers, and alarms due to changes in the operational database or device configurations. This SCADA hijack approach was employed at Kyivoblenergo, whereas the phantom mouse in the previous section was used at Prykarpattiaoblenergo and Chernivtsioblenergo.⁶¹ Therefore, CyOTE analysts assess that this technique occurred at approximately 3:35 PM on 23 December 2015.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, as well as IT Cybersecurity and IT Staff could likely have noticed such observables through an investigation. Operations zone stakeholders would likely have observed unauthorized instructions being issued to endpoints. IT Staff and IT Cybersecurity might observe anomalous VPN traffic.

A total of five observables were identified with the use of the [Unauthorized Command Message technique \(T0855\)](#). Three are assessed as highly perceivable (DMS Client Application Issues Commands Directly to the DMS Server; HMI Reflects Unauthorized Process Change Inputs; and Operational Database Device/Process Alarm Triggered by Event).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Unauthorized Command Message technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.10. REMOTE SERVICES TECHNIQUE (T0886) FOR INITIAL ACCESS

The Remote Services technique (T0886) was used for Initial Access. Adversaries began to access legitimate remote services following credential harvesting using RDP from admin accounts to access the domain controller. At least one oblenergo's firewall allowed adversaries to remote admin out of the environment with a capability native to the victim systems.⁶² This access was used just before attack execution on the afternoon of 23 December.⁶³ The phantom mouse phenomenon, where breakers were opened via HMIs, was enabled via RDP.⁶⁴

Observables associated with this technique include increased RDP traffic to or from unusual or unexpected hosts. IT and OT personnel could very likely observe RDP connections in network traffic or host logs. Logs may reveal unexpected usage of administrator accounts for RDP sessions, as well as changes to the time-of-day access patterns. In addition, logs of affected hosts may contain unusual process data in host logs; unexpected radmin connections may trigger network IDS/IPS alerts. Finally, users of affected hosts might observe commands being entered or unexpected mouse movements.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, as well as IT Cybersecurity and IT Staff would likely have perceived these observables through an investigation.

Of the six observables associated with this technique, five are assessed as highly perceivable (Unusual or Unexpected Hosts Observed in Network Traffic or Host Log Details; Unusual Changes to Time-Of-Day RDP Access Patterns Observed in Network Traffic; Unusual radmin-Associated Process Data in Host Logs; Network IDS/IPS Alerts Associated with radmin Connections; User-Observed Anomalous Mouse Movements and Command Execution).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 24 artifacts could be generated by the Remote Services technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.11. GRAPHICAL USER INTERFACE TECHNIQUE (T0823) FOR EXECUTION

The Graphical User Interface technique (T0823) was used for Execution. This technique is referred to as the phantom mouse. At Prykarpattyaoblenergo and Chernivtsioblenergo, remote admin tools at the OS-level were employed to open breakers.⁶⁵ Adversaries took control of employee workstations and opened breakers across the managed substations between 15:30 and 16:30 local time.⁶⁶ Adversaries targeting the first oblenergo started to open circuit breakers at 15:30 and the cyber attack lasted approximately 60 minutes.⁶⁷ OT Staff noticed the phantom mouse and took a video as it was happening in real-time, and were unable to use their mouse and keyboard to interfere with the attack.^{68,69}

Victims of this technique would likely notice mouse movements on the screen, unauthorized inputs, and a potential inability to utilize the mouse or keyboard. If access to the GUI is enabled via RDP, additional observables would include an increase in RDP traffic associated with remote GUI sessions. Such RDP traffic may also originate from a new or unusual source host, occur at an odd time, or be via a session with an unusual or unexpected user.

CyOTE analysts assess that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff could likely have perceived such observables through an investigation. Operations zone stakeholders would likely have noticed the phantom mouse movements, and IT Staff and IT Cybersecurity might observe anomalous RDP traffic.

A total of seven observables were identified with the use of the [Graphical User Interface technique \(T0823\)](#). Six are likely highly perceivable (Mouse Movement on Screen; HMI Reflects Unauthorized Change Inputs; RDP Traffic from New or Unusual Source Host; RDP Traffic at Odd Hours; Inability to Use Mouse / Keyboard; RDP Sessions from an Unusual or Unexpected User).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 58 artifacts could be generated by the Graphical User Interface technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.12. MANIPULATION OF CONTROL TECHNIQUE (T0831) FOR IMPACT

Adversaries employed the Manipulation of Control technique (T0831) for Impact. Observables associated with an Impact tactic depend upon the business process supported by the Operations Zone. Within the Energy Sector, substation breakers physically changing positions indicate manipulation of control over the electrical power topology.^{70,71} Secondly, unanticipated command traffic to control operational assets are a general observable for this technique that is sector agnostic.

In the case of the December 2015 attack against Ukraine, attackers manipulated physical control processes by sending valid commands to open substation breakers managed by the victim oblenergos. Over a period of approximately 60 minutes, attackers were able to open breakers in at least 57 substations, resulting in widespread power outages.⁷² For Prykarpattyaoblenergo, attackers took between 20 and 30 minutes to inflict an outage.⁷³

CyOTE analysts determined that Engineering and OT Staff would likely observe anomalous device activity and command messages during the attack.

A total of two observables, both of which are assessed as highly perceivable, were identified with the [Manipulation of Control technique \(T0831\)](#) (Breakers Physically Changing Positions; Anomalous Device Activity and Command Messages).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Manipulation of Control technique
Technique Observers	Engineering, OT Staff
Resources	Technique Detection References

3.13. DENIAL OF CONTROL TECHNIQUE (T0813) FOR IMPACT

The Denial of Control technique (T0813) was used for Impact. Adversaries modified user passwords to hamper the ability of the victims to recover and then, during the attack, took control of workstations and locked out operators.^{74,75,76,77} Observables include the inability to control input devices (keyboard, mouse, etc.) to a system.

CyOTE analysts determined that Engineering, OT Staff, and OT Cybersecurity would likely perceive these observables during an attack.

Two observables were associated with the use of the Denial of Control technique (T0813), one of which is assessed as highly perceivable (Inability to Control Input Devices (keyboard, mouse, etc.) to System).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 8 artifacts could be generated by the Denial of Control technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.14. SYSTEM FIRMWARE TECHNIQUE (T0857) FOR IMPAIR PROCESS CONTROL

Adversaries used the Module Firmware technique (T0857) to Impair Process Control. After opening breakers, the adversaries updated firmware for Serial-to-Ethernet communications devices, rendering them inoperable.^{78,79,80} As a result, Engineers at the affected oblenergogs were unable to open breakers to restore power remotely. Instead, Engineers took SCADA systems offline and went to manual mode to restore power, traveling to distribution substations to manually open and close breakers. The Oblenergogs were still operating breakers manually over two months after the attack.⁸¹

Observables associated with this technique include an unauthorized modification made to device firmware, which rendered field devices inoperable. At least 16 substations were disconnected from the control network using the malicious firmware updates. Data transfers directly to field devices, either at an unexpected time or of an unusual size, may be associated with implementations of this technique.

CyOTE analysts determined that Engineering, OT Staff, and OT Cybersecurity would likely perceive these observables during an attack.

A total of four observables were associated with the use of the System Firmware technique (T0857). Two of these observables were assessed as highly perceivable (Unauthorized Modification of Device Firmware; Bricked Device Due to Unsuccessful Firmware Update).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Module Firmware technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.15. BLOCK COMMAND MESSAGE TECHNIQUE (T0803) FOR INHIBIT RESPONSE FUNCTION

The Block Command Message technique (T0803) was used for Inhibit Response Function. During the Ukraine 2015 attack, adversaries severed communications between oblenergo control centers and field devices in substations using two main approaches; the elements of this technique are very similar to those for the Block Reporting Message technique (T0804) in Section 3.16.

First, Engineers at impacted workstations could observe SCADA commands to open breakers being issued but were unable to use their mouse and keyboard to interfere with the attack.^{82,83,84} Second, adversaries updated and corrupted the firmware of Serial-to-Ethernet converters. Once corrupted, these devices would have to be physically replaced to enable breakers to be opened and closed remotely.

When command messages are blocked by corrupting a converter, the following three observables may occur. First, a large file transfer to a converter typically doesn't occur in an OT environment. Second, if the protocol used to upload the malicious firmware is supported by passive network monitoring deep packet inspection, then the firmware upload might be detected and/or extracted. Third, command messages that depend on corrupted protocol converters will not reach source or destination assets. CyOTE analysts determined that Engineering and OT Staff monitoring operational networks and end devices would observe a lack of response to commands even after control of the network was restored.⁸⁵

A total of three observables were associated with the use of the [Block Command Message technique \(T0803\)](#), all of which are assessed to be highly perceivable (File Transfer Destined to Converter; Firmware Upload can be Extracted if the Protocol Used is Supported by Passive Network Monitoring Deep Packet Inspection; Non-Responsive Assets and Command Message Time-Outs).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 9 artifacts could be generated by the Block Command Message technique
Technique Observers	Engineering, OT Staff
Resources	Technique Detection References

3.16. BLOCK REPORTING MESSAGE TECHNIQUE (T0804) FOR INHIBIT RESPONSE FUNCTION

The Block Reporting Message technique (T0804), associated with the adversary's corruption of Serial-to-Ethernet converters, was used to Inhibit Response Function. Reporting messages that depend on corrupted protocol converters will not reach source or destination assets, resulting in message time-outs and non-responsive assets.

During the Ukraine 2015 attack, adversaries severed communications between oblenergo control centers and field devices in substations by updating and corrupting the firmware of Serial-to-Ethernet converters.^{86,87,88,89,90} Operators would have to physically replace these converters to restore communications between field devices and the control center.

CyOTE analysts determined that Engineering and OT Staff monitoring operational networks and end devices would observe unresponsive commands even after control of the network was restored.⁹¹

One observable was identified with the [Block Reporting Message technique \(T0804\)](#) and it is assessed as highly perceivable (Non-Responsive Assets and Reporting Message Time-Outs).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 18 artifacts could be generated by the Block Reporting Message technique
Technique Observers	Engineering, OT Staff
Resources	Technique Detection References

3.17. LOSS OF CONTROL TECHNIQUE (T0827) FOR INHIBIT RESPONSE FUNCTION

Adversaries used the Loss of Control technique (T0827) to Inhibit Response Function. Although this technique may apply to a variety of systems, observables associated with the Ukraine 2015 attack include inaccessible systems, systems not booting, and error messages or application crashes due to missing critical files. Binaries extracted from network traffic or a host that match Killdisk file signatures in Yara or other signature tools provide an additional observable.

On 23 December, adversaries implemented the Loss of Control Technique with a variant of KillDisk featuring a timer, scheduling outages to UPS systems, and uploading firmware to disable Serial-to-Ethernet converters at substations.⁹² The overall result was a loss of system control for enterprise systems such as finance and human resources via KillDisk, a loss of telephon connectivity at Prykarpattyaoblenergo via scheduling UPS outages, and a loss of SCADA so severe that operators took those systems offline.⁹³

CyOTE analysts determined that Engineering, OT Staff, OT Cybersecurity, Management, Support Staff, IT Cybersecurity, and IT Staff very likely would have observed elements of this technique.

Analysts identified four observables, all assessed to be highly perceivable, with the use of the [Loss of Control technique \(T0827\)](#) (Extraction of Binary That Matches Killdisk File Signatures in Yara or Other File Signature Tools from Extracted Files from Network Traffic or From a Host; Systems Remotely Inaccessible/Non-Responding; System Does Not Boot; Error Messages or Application Crashes Associated with Missing Critical Files).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 9 artifacts could be generated by the Loss of Control technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, Management, Support Staff, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.18. DENIAL OF SERVICE TECHNIQUE (T0814) FOR INHIBIT RESPONSE FUNCTION

The Denial of Service technique (T0814) was used for Inhibit Response Function. Several observables are associated with a TDoS attack, which was employed during the Ukraine 2015 attack. During a TDoS, calls flood a targeted call center, resulting in a large wait queue. The volume of phone calls may result in high utilization of telephone infrastructure, denying users the ability to dial an impacted phone number, or the infrastructure itself may fail.

During the Ukraine 2015 attack, adversaries launched a TDoS attack on call centers to disrupt operations and restoration efforts of Prykarpattiaoblenergo and Kyivoblenergo. Kyivoblenergo was unable to receive calls about the location of outages due to a flood of bogus, automated telephone calls from foreign phone numbers. This degraded its ability to respond quickly because the oblenergo had no situational awareness without inputs from the HMIs and was unable to receive calls from customers about where outages had occurred. The TDoS attack compounded the effect on the oblenergos' response capability after adversaries took the SCADA systems offline.⁹⁴ The adversaries further degraded communications through a scheduled UPS outage (see Section 3.21) that impacted a telephony communications server at one oblenergo.⁹⁵

CyOTE analysts determined that Engineering, OT Staff, IT Cybersecurity, and IT Staff would all notice observables associated with this technique.

A total of five observables, all of which are assessed as highly perceivable, were identified with the use of the [Denial-of-Service technique \(T0814\)](#) (Flood of Calls to Operator Telephone Call Center; High Call Center Wait Queue; Inability to Dial Impacted Phone Number; High Utilization of Failure of Telephone Infrastructure; Unavailability of Telephone Services).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 14 artifacts could be generated by Denial of Service technique
Technique Observers	Engineering, OT Staff, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.19. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION

The Service Stop technique (T0881) was used for Inhibit Response Function. During the Ukraine 2015 attack, adversary behavior targeting oblenergo systems stopped critical functions in the cyber and physical domains.⁹⁶ In the physical domain, attackers stopped electrical service for an estimated 225,000 customers by taking offline multiple substations across three oblenergos. Adversaries targeted Serial-to-Ethernet converters, rendering them unrecoverable by the manufacturer, to deny operators the ability to open and close substation breakers remotely.^{97,98}

In addition, using KillDisk, adversaries overwrote the MBRs of multiple machines in enterprise and operational environments, impacting a variety of business functions.⁹⁹ Filenames such as svchost.exe, tsks.exe, crab.exe, and virus_ololo.dat are correlated with the implementation of this technique in the Ukraine 2015 attack.

Stopped network services might result in an increase in failed network connections observed in network traffic (e.g., TCP RST). In addition, inoperable systems might result in an increase in network messages associated with an unknown or unavailable host (e.g., ICMP host unreachable). Due to unavailable systems, operators might notice that process functions are not working as expected or that systems cannot be accessed.

CyOTE analysts determined that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff would likely perceive the effects of this technique during an investigation.

A total of four observables were identified, all assessed to be highly perceivable, with the use of the [Service Stop technique \(T0881\)](#) (Stopped Services Might Result in an Increase in Failed Network Connections Observed in Network Traffic; Inoperable Systems Might Result in an Increase in ICMP Host Unreachable or Other Network Messages Associated with an Unknown or Unavailable Host; Operators Process Functions are not Working as Expected or Systems Cannot Be Accessed; Filenames Specific to the 2015 Campaign Malware).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by the Service Stop technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, IT Staff
Resources	Technique Detection References

3.20. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION

Adversary malware used the Data Destruction technique (T0809) for Inhibit Response Function. This technique inhibits recovery attempts made by the victim through deletion of critical OS files. In general, effects of this technique include file signatures from files extracted from network data or recovered from an endpoint that contains KillDisk signature matches. Users might be unable to login to victim systems and systems might crash or throw error messages.

During the Ukraine 2015 attack, adversaries used a specific variant of Win32/Killdisk.NBB.¹⁰⁰ This variant included a command line argument for a time delay, the ability to delete Windows Event Logs, and a targeted list of document extensions.¹⁰¹ Targets were mainly enterprise networks and included servers and hosts used by management, human resources, and finance staff; some Remote Terminal Units (RTUs) in the OT environment were also impacted.¹⁰²

CyOTE analysts determined that Engineering, OT Staff, OT Cybersecurity, IT Cybersecurity, and IT Staff would observe effects of this technique.

A total of five observables were identified with the use of the [Data Destruction technique \(T0809\)](#). These observables almost certainly would have been visible within enterprise and operational environments, as the technique renders machines unusable by overwriting their MBRs.

IT Staff, IT Cybersecurity, OT Staff, and Engineering would observe a loss of connectivity to hosts, non-responsive hosts, and missing files.

CyOTE analysts assessed two of the five observables as highly perceivable (Users Unable to Login to Systems; Systems Crashing or Throwing Error Messages Associated with Critical Errors).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 27 artifacts could be generated by the Data Destruction technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.21. DEVICE RESTART/SHUTDOWN TECHNIQUE (T0816) FOR INHIBIT RESPONSE FUNCTION

The Device Restart/Shutdown technique (T0816) was used for Inhibit Response Function. Devices are unavailable while restarting or after shutting down. These systems remained inaccessible and did not respond even if restarted after their MBRs were deleted. System log entries denoting system shutdown will be present on the host until deleted and will be visible if central logging is enabled.

During the Ukraine 2015 attack, adversaries inhibited the oblenergog's response following the power outage by scheduling outages to systems infected with KillDisk malware and compromised UPS systems that could provide backup power.¹⁰³ When KillDisk runs, after targeting the MBR on a device it restarts the host, rendering it inoperable.¹⁰⁴ UPS outages were scheduled using remote management interfaces before the first breakers were opened.¹⁰⁵

As with the TDoS attack, this technique may have been an alternative approach to deny telephone communications to Prykarpattyaoblenergo, where the UPS of the Private Branch Exchange (PBX) was shut down.^{106,107} Additional impacts included the backup power to oblenergo control centers.¹⁰⁸

CyOTE analysts determined that Engineering, OT Staff, OT Cybersecurity, Management, Support Staff, IT Cybersecurity, and IT Staff would have observed the impact of this technique.

A total of three observables, all of which are assessed as highly perceivable, were identified with the use of the [Device Restart/Shutdown technique \(T0816\)](#) (Inability to Access Remote Services on Device Impacted by Restart/Shutdown Attack; Systems Inaccessible/Non-Responding; System Log Entries Denoting System Was Restarted/Shutdown).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 17 artifacts could be generated by the Device Restart/Shutdown technique
Technique Observers	Engineering, OT Staff, OT Cybersecurity, Management, Support Staff, IT Cybersecurity, IT Staff
Resources	Technique Detection References

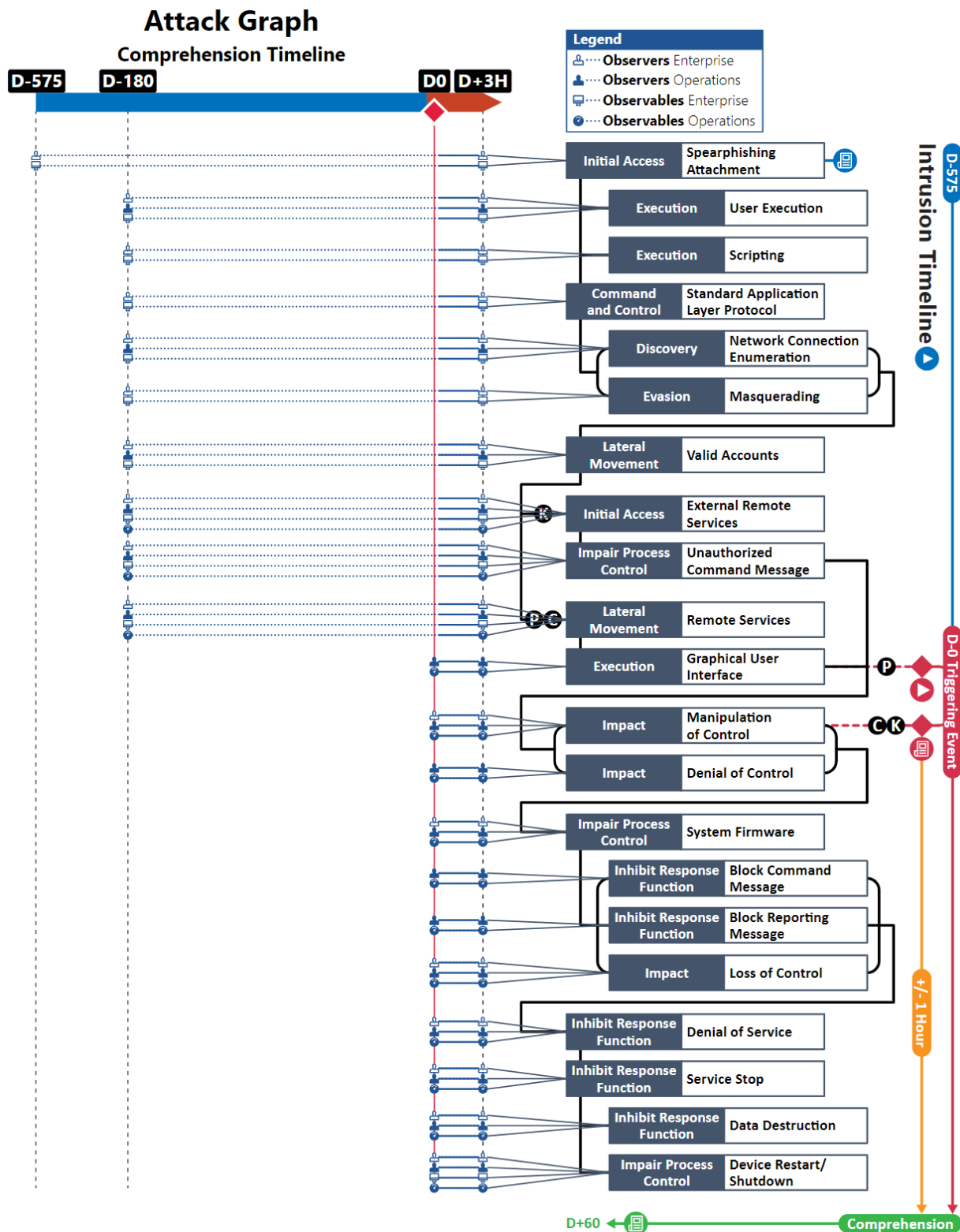


Figure 3. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

Observables Associated with Spearphishing Attachment Technique (T0865)	
Observable 1	Anomalous Outbound Network Connections/Downloads Over Http(S)
Observable 2	Object Linking and Embedding (OLE) Objects in Malicious Word/Excel Docs
Observable 3	Додаток1.xls and Associated Components: "vba_macro.exe". In Some Cases, Associated Components Included a Dropper Installer "DropbearRun.vbs"
Observable 4	\$RR143TB.doc and Associated Components: "vba_macro.exe". In Some Cases, Associated Components Included a Dropper Installer "DropbearRun.vbs"
Observable 5	Download Of Pyinstaller Compiled Gcat Binary
Observable 6	Unusual SMTP Traffic to Google or Other Web Mail Provider. If the User Primarily Uses a Browser to Access Email, Gcat Will Generate Unusual SMTP
Observable 7	File Signatures in Files Extracted from Network Traffic or Host Data Matching BlackEnergy or Gcat Signatures (Example: Yara)

Observables Associated with User Execution Technique (T0863)	
Observable 1	Office Documents (Word, Excel) Associated with Network Communications, Command Prompt Processes or Other Suspicious Child Process
Observable 2	HTTP or SMB Protocols Containing Known Malicious Signatures for C2 or Malware Propagation. This Might Be Seen in Network Signatures or File Signatures Associated with Extracted Files
Observable 3	Suspicious Emails with Spoofed Sender Addresses and Attachments that Prompt Users To Enable Macros When Downloaded

Observables Associated with Scripting Technique (T0853)	
Observable 1	Office Document (Word, Excel) Containing OLE or Other Scripting Object Such as "DropbearRun.vbs"
Observable 2	Network Traffic Associated with Microsoft Office (Word, Excel) Processes
Observable 3	Filenames Specific To 2015 Campaign Included Додаток1.xls, \$RR143TB.doc, VBA_macro.exe, CPLEXE.EXE (Original Name), MS-IME (Internal Name), virus_04.exe, vba_macro, icshextobin.exe, BlackEnergy.exe, 1.exe

Observables Associated with Standard Application Layer Protocol Technique (T0869)	
Observable 1	Anomalous Outbound Network Connections/Downloads Over HTTP(S)
Observable 2	Office Document (Word, Excel) Containing OLE or Other Scripting Object as Parent Process Associated to HTTP Comms to External IP
Observable 3	HTTP Request to IP Directly with No Associated DNS Request. This May Include an HTTP POST Request with the Following Fields: Id=[bot_id_sha1]&bid=[base64_encoded_build_id]&nm=[x]&cn=[y]&num=[z]
Observable 4	SMTP Connections to Unusual or Unexpected Mail Services

Observables Associated with Network Connection Enumeration Technique (T0840)	
Observable 1	Network Traffic with Scanning Behavior (Sequential Ports, High Destination to Source Port Communication Ratio, High Quantity of Partial Connections) Might Be Noted from Unusual Systems
Observable 2	Filenames specific to the 2015 Campaign Included: VS.dll, SCAN.dll, DC.dll, BS.dll
Observable 3	Anomalous Use of MS Sysinternals PsExec Tool

Observables Associated with Masquerading Technique (T0849)	
Observable 1	Executable with Common Name Outside of Expected Folder. In this Event, the Malicious svchost.exe Malware File Was Located at C:\Windows\Temp\Syslog\ While the Legitimate svchost.exe Is Stored at C:\Windows\System32\
Observable 2	Unsigned or Not Valid Signature on Operating System or Application File
Observable 3	Unusual Publisher Signature on Operating System or Application File
Observable 4	Valid Signature but Invalid or Unknown File Hash on Operating System or Application File
Observable 5	Unusual Imports or Exports on Operating System or Application File
Observable 6	Scanner Tool Executable Renamed to svchost.exe

Observables Associated with Valid Accounts Technique (T0859)	
Observable 1	Suspicious or Unusual Interactive Successful Logins to Domain-Joined Assets Observed in the Domain Controller or Target Machine's Windows Event Logs Security Log (Event 4624)
Observable 2	Suspicious or Unusual Interactive Unsuccessful Logins to Domain-Joined Assets Observed in the Domain Controller or Target Machine's Windows Event Logs Security Log (Event 4625)
Observable 3	SMB Traffic Associated with PsExec User Account Compromised. This Might Be Done with Either Elevated or User Level Accounts
Observable 4	Unusual User Account Logins, Particularly Admin Accounts Across Network Services Such as SMB
Observable 5	Psexec Process Artifacts on Destination System Associated with PSEXESVC.Exe
Observable 6	Movement of Suspicious Files Over SMB or Other Admin Shares
Observable 7	Filenames Specific to the 2015 Campaign Included: PS.dll, SI.dll, KI.dll
Observable 8	Event ID 7045 and 4697

Observables Associated with External Remote Services Technique (T0822)	
Observable 1	VPN Application Logs Show Anomalous Account Activity

Observables Associated with External Remote Services Technique (T0822)

Observable 2	VPN Logon Session Metadata Shows Use of Compromised Acct
Observable 3	Unusual Radmin Associated Process Data in Host Logs
Observable 4	Network IDS/IPS Alerts Associated with Radmin Connections (Example: Snort Rule)

Observables Associated with Unauthorized Command Message Technique (T0855)

Observable 1	Anomalous VPN Connections
Observable 2	DMS Client Application Issuing Commands Directly to DMS Server (Kyivoblenergo)
Observable 3	HMI Reflects Unauthorized Process Change Inputs via radmin (Prykarpattya, Chernivtsi)
Observable 4	PCAP Shows Unauthorized Instructions Sent to Field Controllers
Observable 5	Operational Database Device/Process Alarm Triggered by Event

Observables Associated with Remote Services Technique (T0886)

Observable 1	Increased RDP Traffic to Or from Unusual or Unexpected Hosts Observed in Network Traffic or Host Logs Details
Observable 2	Unexpected Usage of Administrator Accounts for RDP Sessions
Observable 3	Unusual Changes to Time-Of-Day RDP Access Patterns Observed in Network Traffic
Observable 4	Unusual radmin Associated Process Data in Host Logs
Observable 5	Network IDS/IPS Alerts Associated with Radmin Connections (Example: Snort Rule)
Observable 6	Users Observe Commands Being Entered and Mouse Being Moved

Observables Associated with Graphical User Interface Technique (T0823)

Observable 1	Mouse Movement on Screen
Observable 2	HMI Reflects Unauthorized Change Inputs
Observable 3	Increase in RDP Traffic Associated with Remote GUI Sessions
Observable 4	RDP Traffic from New or Unusual Source Host
Observable 5	RDP Traffic at Odd Hours
Observable 6	RDP Sessions from Unusual or Unexpected User
Observable 7	Inability of Users to Utilize Mouse or Keyboard

Observables Associated with Manipulation of Control Technique (T0831)

Observable 1	Breakers Physically Changing Positions
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Observables Associated with Manipulation of Control Technique (T0831)

Observable 2	Network Command Traffic Shows Unauthorized Commands Sent to Breakers
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Observables Associated with Denial of Control Technique (T0813)

Observable 1	Inability to Control Input Devices (Keyboard, Mouse, etc.) to System
Observable 2	Modified User Passwords to Hamper Recovery Ability

Observables Associated with System Firmware Technique (T0857)

Observable 1	Unauthorized Modification Made to Device Firmware
Observable 2	Large File or Data Transfer Directly to Convertor
Observable 3	Bricked Device Due to Unsuccessful Firmware Update
Observable 4	Control Center Systems Unable to Connect to Substation Devices

Observables Associated with Block Command Message Technique (T0803)

Observable 1	File Transfer Destined to Converter. This Doesn't Typically Occur in OT Environment
Observable 2	Firmware Upload Can Be Extracted if the Protocol Used is Supported by Passive Network Monitoring Deep Packet Inspection
Observable 3	Non-Responsive Assets and Command Message Time-Outs (as Command Messages that Depend on Corrupted Protocol Converters Will not Reach Source or Destination Assets)

Observables Associated with Block Reporting Message Technique (T0804)

Observable 1	Non-Responsive Assets and Reporting Message Time-Outs (as Reporting Messages that Depend on Corrupted Protocol Converters Will not Reach Source or Destination Assets)
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Observables Associated with Loss of Control Technique (T0827)

Observable 1	Extraction of Binary That Matches Killdisk File Signatures in Yara or Other File Signature Tools from Extracted Files from Network Traffic or From a Host
Observable 2	Systems Remotely Inaccessible/Non-Responding
Observable 3	System Does not Boot
Observable 4	Error Messages or Application Crashes Associated with Missing Critical Files

Observables Associated with Denial of Service Technique (T0814)

Observable 1	Flood of Calls to Operator Telephone Call Center
Observable 2	High Call Center Wait Queue

Observables Associated with Denial of Service Technique (T0814)

Observable 3	Inability to Dial Impacted Phone Number
Observable 4	High Utilization or Failure of Telephone Infrastructure
Observable 5	Unavailability of Telephone Services.

Observables Associated with Service Stop Technique (T0881)

Observable 1	Stopped Services Might Result in an Increase in Failed Network Connections Observed in Network Traffic (TCP Resets, etc.)
Observable 2	Inoperable Systems Might Result in an Increase in ICMP Host Unreachable or Other Network Messages Associated with an Unknown or Unavailable Host
Observable 3	Operators Might Notice That Process Functions Are not Working as Expected or That Systems Cannot Be Accessed
Observable 4	Filenames Specific to the 2015 Campaign Malware Included: svchost.exe, tsk.exe, danger, Ukrainian.bin.exe, crab.exe, ololo.exe, ololo 2.exe, ololo.txt, trololo.exe, 123.txt, virus_ololo.dat

Observables Associated with Data Destruction Technique (T0809)

Observable 1	File Signatures from Files Extracted from Network Data or Recovered from an Endpoint Might Contain Killdisk Signature Matches (Example Yara)
Observable 2	Users Unable to Login to Systems
Observable 3	Systems Crash or Throw Error Messages
Observable 4	Absence of System Log Data
Observable 5	Absence of MBR Data

Observables Associated with Device Restart/Shutdown Technique (T0816)

Observable 1	Inability to Access Remote Services on Device Impacted by Restart/Shutdown Attack
Observable 2	Systems Inaccessible/Non-Responding
Observable 3	System Log Entries Denoting System Was Restarted / Shutdown

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Spearphishing Attachment Technique (T0865)	
Artifact 1	Mismatch MIME and Attachment File Extension
Artifact 2	Email Sender Address
Artifact 3	Email Message
Artifact 4	Email Receiver
Artifact 5	Email Receiver Name
Artifact 6	Email Receiver Domain
Artifact 7	Email Receiver Address
Artifact 8	Enable Macros Pop-Up
Artifact 9	Email Application Log File
Artifact 10	Email Unified Audit Log File
Artifact 11	Email Service Name
Artifact 12	Suspicious Email Message Content
Artifact 13	Operating System Service Creation
Artifact 14	Email .pst File
Artifact 15	Email .ost File
Artifact 16	Simple Mail Transfer Protocol (SMTP) Traffic
Artifact 17	Mail Transfer Agent Logs
Artifact 18	Email Parent Process
Artifact 19	Mail Transfer Agent Logs
Artifact 20	Email Parent Process
Artifact 21	Email Domain Name System (DNS) Traffic
Artifact 22	Email Domain Name System (DNS) Event
Artifact 23	File Attachment Warning Prompt
Artifact 24	Email Timestamp
Artifact 25	Email Attachment
Artifact 26	Email Attachment File Type
Artifact 27	Email Header
Artifact 28	Email Sender Name
Artifact 29	Email Sender IP Address
Artifact 30	Email Sender Domain

Artifacts Associated with User Execution Technique (T0863)	
Artifact 1	Application Log

Artifacts Associated with User Execution Technique (T0863)	
Artifact 2	Prefetch Files
Artifact 3	System Log
Artifact 4	Registry Modification
Artifact 5	File Modifications
Artifact 6	File Renaming
Artifact 7	System Patches Installed
Artifact 8	Files Opening
Artifact 9	File Signature Validation
Artifact 10	Installers Created
Artifact 11	Process Termination
Artifact 12	File Creation
Artifact 13	Service Termination
Artifact 14	File Changes
Artifact 15	User Account Modification
Artifact 16	Increased ICMP Traffic (Network Scanning)
Artifact 17	File Execution
Artifact 18	Network Traffic Changes
Artifact 19	Process Creation
Artifact 20	Network Connection Creation
Artifact 21	Command Execution
Artifact 22	Application Log Content
Artifact 23	Application Installation

Artifacts Associated with Scripting Technique (T0853)	
Artifact 1	Files Dropped into Directory
Artifact 2	System Event Log Creation
Artifact 3	OS Timeline Event
Artifact 4	Startup Menu Modification
Artifact 5	System Processes Created
Artifact 6	Windows API Event Log
Artifact 7	Executable Files
Artifact 8	Prefetch Files Created
Artifact 9	External Network Connections
Artifact 10	Network Services Created

Artifacts Associated with Scripting Technique (T0853)

Artifact 11	Registry Modifications
Artifact 12	OS Service Installation

Artifacts Associated with Standard Application Layer Protocol Technique (T0869)

Artifact 1	External Network Connections
Artifact 2	DNS Autonomous System Number
Artifact 3	Increase In the Number of External Connections
Artifact 4	Network Content Metadata
Artifact 5	Network Connection Times
Artifact 6	HTTP Traffic Port
Artifact 7	DNS Traffic Port
Artifact 8	SMB Traffic Port
Artifact 9	HTTPS Traffic Port
Artifact 10	RDP Traffic Port
Artifact 11	HTTP Post Request
Artifact 12	External IP Addresses

Artifacts Associated with Network Connection Enumeration Technique (T0840)

Artifact 1	Common Network Traffic
Artifact 2	Polling Network Traffic from Abnormal IP Sender Addresses
Artifact 3	NETBIOS Name Services Port
Artifact 4	LDAP Port
Artifact 5	Active Directory Calls
Artifact 6	Email Server Calls
Artifact 7	SMTP Port 25 Traffic
Artifact 8	DNS Lookup Queries
Artifact 9	ARP Scans
Artifact 10	TCP Connect Scan
Artifact 11	TCP SYN Scans
Artifact 12	Industrial Network Traffic
Artifact 13	TCP FIN Scans
Artifact 14	TCP Reverse Ident Scan
Artifact 15	TCP XMAS Scan
Artifact 16	TCP ACK Scan

Artifacts Associated with Network Connection Enumeration Technique (T0840)

Artifact 17	VNC Port 5900 Calls
Artifact 18	Protocol Content Enumeration
Artifact 19	Protocol Header Enumeration
Artifact 20	Recurring Protocol SYN Traffic
Artifact 21	Sequential Protocol SYN Traffic
Artifact 22	Statistical Anomalies In Network Traffic
Artifact 23	Echo Port 8 Traffic
Artifact 24	Device Failure
Artifact 25	Device Reboot
Artifact 26	Bandwidth Degradation
Artifact 27	Host Recent Connection Logs
Artifact 28	ICMP Port 7 Traffic
Artifact 29	SNMP Port 162 Traffic
Artifact 30	SNMP Port 161 Traffic
Artifact 31	Command Line Dialog Box Open
Artifact 32	Operating System Queries
Artifact 33	DNS Port 53 Zone Transfers

Artifacts Associated with Masquerading Technique (T0849)

Artifact 1	File Creation with Common Name
Artifact 2	Additional File Directories Created
Artifact 3	Scheduled Job Modification
Artifact 4	Service Creation
Artifact 5	Services Metadata
Artifact 6	Leetspeak User Metadata
Artifact 7	Common Application with Non-Native Child Processes
Artifact 8	Process Metadata Changes
Artifact 9	Command Line Execution
Artifact 10	File Modification
Artifact 11	Warez Application Use
Artifact 12	Leetspeak File Creation
Artifact 13	Applications Causing Unintended Actions
Artifact 14	Additional Functionality in Applications

Artifacts Associated with Valid Accounts Technique (T0859)	
Artifact 1	Logons
Artifact 2	Default Credential Use
Artifact 3	Application Log
Artifact 4	Domain Permission Requests
Artifact 5	Permission Elevation Requests
Artifact 6	Application Use Times
Artifact 7	Configuration Changes
Artifact 8	Prefetch Files Created After Execution
Artifact 9	Logon Session Creation
Artifact 10	User Account Creation
Artifact 11	Authentication Creation
Artifact 12	System Logs
Artifact 13	Successful Logon Event
Artifact 14	Failed Logons Event
Artifact 15	Logon Timestamp
Artifact 16	Logon Type Entry

Artifacts Associated with External Remote Services Technique (T0822)	
Artifact 1	Remote Services Protocols
Artifact 2	VPN Connections
Artifact 3	Remote Vendor Connections
Artifact 4	Session Authentication
Artifact 5	Failed Logons Event
Artifact 6	Session Timestamp
Artifact 7	Logon Event Type
Artifact 8	Remote Session Key
Artifact 9	System Registry Network Interfaces
Artifact 10	Remote Services Logon
Artifact 11	TLS Certificate
Artifact 12	Session Logoff Event
Artifact 13	Domain Controller Log
Artifact 14	User Account Name
Artifact 15	User Client Address
Artifact 16	Security Account Manager Registry Entries

Artifacts Associated with External Remote Services Technique (T0822)	
Artifact 17	Dialog Box Pop-Up
Artifact 18	Mouse Movement
Artifact 19	Command Prompt Window Opened
Artifact 20	Security Account Manager Registry Password Hashes
Artifact 21	External IP Address
Artifact 22	User Account Creation
Artifact 23	User Privileges Change
Artifact 24	Blocked Incoming Connections Event
Artifact 25	Blocked Incoming Packet Event
Artifact 26	Encrypted Network Traffic

Artifacts Associated with Unauthorized Command Message Technique (T0855)	
Artifact 1	Protocol Specific Command Packet
Artifact 2	Machine State Change
Artifact 3	Process Restart
Artifact 4	Process Failure
Artifact 5	Network Resets
Artifact 6	Protocol Metadata Change
Artifact 7	Process Timing Change
Artifact 8	Process Logic Change
Artifact 9	MAC Addresses
Artifact 10	IP Addresses
Artifact 11	Operational Application Log
Artifact 12	Operational Data Created
Artifact 13	Process Alarm
Artifact 14	Process Alarm Event
Artifact 15	Application Level I/O Manipulation
Artifact 16	OS Level I/O Manipulation

Artifacts Associated with Remote Services Technique (T0886)	
Artifact 1	Remote Client Connection
Artifact 2	Logon Event
Artifact 3	Logoff
Artifact 4	Logoff Event

Artifacts Associated with Remote Services Technique (T0886)	
Artifact 5	Registry Changes
Artifact 6	Registry Connection Change
Artifact 7	Mouse Movement
Artifact 8	Unexpected
Artifact 9	Desktop Prompt Windows Created
Artifact 10	Session Cache
Artifact 11	Application Log
Artifact 12	RDP Traffic
Artifact 13	System Log Event
Artifact 14	Authentication Logs
Artifact 15	GUI Modifications
Artifact 16	Data File Size In Network Content
Artifact 17	File Movement
Artifact 18	MSSQL Traffic 1433 Port
Artifact 19	SSH Traffic
Artifact 20	SMB Traffic
Artifact 21	VNC Traffic
Artifact 22	Process Creation
Artifact 23	Remote Session Creation Timestamp
Artifact 24	Network Traffic Content Creation

Artifacts Associated with Graphical User Interface Technique (T0823)	
Artifact 1	Remote Services Protocols
Artifact 2	VPN Connections
Artifact 3	Remote Vendor Connections
Artifact 4	Session Authentication
Artifact 5	Failed Logons Event
Artifact 6	Session Timestamp
Artifact 7	Logon Event Type
Artifact 8	Remote Session Key
Artifact 9	System Registry Network Interfaces
Artifact 10	Remote Services Logon
Artifact 11	TLS Certificate
Artifact 12	Session Logoff Event

Artifacts Associated with Graphical User Interface Technique (T0823)	
Artifact 13	Domain Controller Log
Artifact 14	User Account Name
Artifact 15	User Client Address
Artifact 16	Security Account Manager Registry Entries
Artifact 17	Dialog Box Pop-Up
Artifact 18	Mouse Movement
Artifact 19	Command Prompt Window Opened
Artifact 20	Security Account Manager Registry Password Hashes
Artifact 21	External IP Address
Artifact 22	User Account Creation
Artifact 23	User Privileges Change
Artifact 24	Blocked Incoming Connections Event
Artifact 25	Blocked Incoming Packet Event
Artifact 26	Encrypted Network Traffic
Artifact 27	External MAC Address
Artifact 28	Event Log Creation
Artifact 29	Logon Event
Artifact 30	SMB Port
Artifact 31	RDP Port
Artifact 32	SSH Port
Artifact 33	Mouse Movement
Artifact 34	Cursor Movement
Artifact 35	Keyboard Entries
Artifact 36	Application Execution via Input Devices
Artifact 37	Prefetch Files Created
Artifact 38	VNC Connections
Artifact 39	RDP Connections
Artifact 40	Program Executions
Artifact 41	Code Injections
Artifact 42	Host-Screen Adjustments
Artifact 43	Screen Resolution Changes
Artifact 44	SSH Connections
Artifact 45	Process Creations
Artifact 46	Service Creation
Artifact 47	Service Modification

Artifacts Associated with Graphical User Interface Technique (T0823)

Artifact 48	Process Input Changes
Artifact 49	JUMPLIST Creation
Artifact 50	SHELLBAG Creation
Artifact 51	System Resource Use Management Changes
Artifact 52	Network Connection Durations
Artifact 53	Changes in Bytes Sent and Received
Artifact 54	Increase CPU Cycles
Artifact 55	Host System Crash
Artifact 56	Application Usage Increase
Artifact 57	Network Bandwidth Changes
Artifact 58	Remote Client Execution

Artifacts Associated with Manipulation of Control Technique (T0831)

Artifact 1	HMI Input Manipulation
Artifact 2	Command Execution
Artifact 3	Event Log Creation
Artifact 4	Application Log Event
Artifact 5	Altered Command Sequences
Artifact 6	Application File Modification
Artifact 7	Operational Data Modification
Artifact 8	I/O Modification
Artifact 9	Engineering Workstation Mouse Movement
Artifact 10	Controller Set Point Change
Artifact 11	Controller Parameter Change
Artifact 12	Controller Tag Change
Artifact 13	Process State Change
Artifact 14	Process Shutdown
Artifact 15	Process Restart
Artifact 16	Process Initiated

Artifacts Associated with Denial of Control Technique (T0813)

Artifact 1	Input Failure
Artifact 2	Increased Network Packet Delivery
Artifact 3	Process Failure

Artifacts Associated with Denial of Control Technique (T0813)

Artifact 4	Process Reboot
Artifact 5	Serial Communication Failure
Artifact 6	Network Ports Opened
Artifact 7	Network Ports Closed
Artifact 8	Process Nonresponsive

Artifacts Associated with Module Firmware Technique (T0857)

Artifact 1	Firmware Update Request
Artifact 2	Device Restart
Artifact 3	Device Shutdown
Artifact 4	Protection Relay Modification
Artifact 5	Protection Relay Status Change
Artifact 6	User Account Creation
Artifact 7	Administrator Account Use
Artifact 8	Device Authentication Event
Artifact 9	Network Traffic
Artifact 10	Operations Device Failure
Artifact 11	External Network Connections to Operational Devices
Artifact 12	External Network Exchange of Data
Artifact 13	I/O Server Modifications
Artifact 14	Device Status Change
Artifact 15	Device Tag Manipulation
Artifact 16	Code Execution

Artifacts Associated with Block Command Message Technique (T0803)

Artifact 1	New Network Connections Created
Artifact 2	Supervisory Application Log Event
Artifact 3	Supervisory Application Log Failure
Artifact 4	Application Processes Terminated
Artifact 5	Operational Data Mismatch Process Physical State
Artifact 6	Historian Data Missing
Artifact 7	Historian Data Query Failure
Artifact 8	Operational Database Event Alarms
Artifact 9	Input Failed to Change Operations Device

Artifacts Associated with Block Reporting Message Technique (T0804)	
Artifact 1	Application Modification
Artifact 2	Physical Process Changes Without Data Received
Artifact 3	Conflicting Device Status Reports
Artifact 4	I/O Values Mismatched with Process Current State
Artifact 5	Delayed Operational Process Status Change
Artifact 6	Application Log Event Absent
Artifact 7	Historian Database Missing Data
Artifact 8	I/O Server Nonresponsive
Artifact 9	Real-Time Operational Data Missing
Artifact 10	Supervisory Application Logs Mismatch Current State
Artifact 11	Process Status Modification
Artifact 12	Network Traffic Changes
Artifact 13	Network Connections Creation
Artifact 14	Operational Device Failure
Artifact 15	Operational Database Data Modification
Artifact 16	Operational Database Configuration Change
Artifact 17	Operational Process Termination
Artifact 18	Operational Process Alarm Failures

Artifacts Associated with Loss of Control Technique (T0827)	
Artifact 1	Process Alarms
Artifact 2	Machine State Change
Artifact 3	Configuration Change
Artifact 4	Set Point Failure
Artifact 5	Service Request Increases
Artifact 6	Runaway Conditions
Artifact 7	Failed Input Commands
Artifact 8	Process Environment Changes
Artifact 9	Device Failure
Artifact 10	Network Connection Loss
Artifact 11	Unresponsive I/O Conditions
Artifact 12	Process Failure
Artifact 13	Repeated Maintenance Reports

Artifacts Associated with Denial of Service Technique (T0814)	
Artifact 1	Application Log
Artifact 2	TDS Traffic Increase Port
Artifact 3	Increase Industrial Protocol Exceptions
Artifact 4	Low Resources Warning
Artifact 5	Ransom Notice
Artifact 6	Services Failure
Artifact 7	Network Traffic Connection Increase
Artifact 8	IP Addresses
Artifact 9	MAC Addresses
Artifact 10	External Network Connections
Artifact 11	Process Performance Degrades
Artifact 12	Operational Data Corruption
Artifact 13	Application Failure
Artifact 14	ICMP Echo Port 7 Traffic Increase

Artifacts Associated with Service Stop Technique (T0881)	
Artifact 1	Process Failure
Artifact 2	Alarm Event
Artifact 3	Internal System Logs
Artifact 4	Application Error Messages
Artifact 5	Process Error Messages
Artifact 6	Application Service Stop
Artifact 7	OS Service Crash
Artifact 8	System Event Logs
Artifact 9	Application Event Logs
Artifact 10	OS API Call
Artifact 11	Command Line System Argument
Artifact 12	System Resource Usage Manager Application Usage Change
Artifact 13	Registry Change HKLM\SYSTEM\CURRENTCONTROLSET\SERVICES

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 1	Program Execution
Artifact 2	Telnet Port

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 3	SFTP Port
Artifact 4	FTPS Port
Artifact 5	SMB Port
Artifact 6	HTTP Port
Artifact 7	HTTPS Port
Artifact 8	Command Line Arguments
Artifact 9	SCP Port
Artifact 10	Memory Corruption
Artifact 11	Files Moved to Recycle Bin
Artifact 12	Non-Native Files
Artifact 13	Transient Device Connections
Artifact 14	External Network Connections
Artifact 15	Local Network Connections
Artifact 16	Host System Reboot Failure
Artifact 17	Process Logic Failure
Artifact 18	Event Log Creation
Artifact 19	System Call
Artifact 20	System Application Interruption
Artifact 21	Device Failure
Artifact 22	Recovery Attempt Failure
Artifact 23	File Encryptions
Artifact 24	Missing Files
Artifact 25	Use of File Transfer Protocols
Artifact 26	FTP Port
Artifact 27	TFTP Port

Artifacts Associated with Device Restart/Shutdown Technique (T0816)	
Artifact 1	Process Application Event
Artifact 2	Process Environmental Changes
Artifact 3	Loss of Network Connection
Artifact 4	Network Command Packets
Artifact 5	Reboot Screen
Artifact 6	Blue Screen
Artifact 7	Significant Operational Data Changes

Artifacts Associated with Device Restart/Shutdown Technique (T0816)	
Artifact 8	Logon Events
Artifact 9	Logoff Events
Artifact 10	Process Failure
Artifact 11	Hardware Failure
Artifact 12	Command Prompt Opened
Artifact 13	Unauthorized Input
Artifact 14	Memory Corruption
Artifact 15	Process Alarm
Artifact 16	External Network Connections
Artifact 17	Local Network Connections

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	• Security Tester
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

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¹⁰⁵ [E-ISAC and SANS | Robert M. Lee, Michael J. Assante, Tim Conway | “Analysis of the Cyber Attack on the Ukrainian Power Grid – Defense Use Case” | <https://isij.eu/article/analysis-cyber-attack-ukrainian-power-grid-defense-use-case> | 18 March 2016 | Accessed on 7 June 2022 | This source is publicly available information and does not contain classification markings]

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